

Automated Formulation & Wood Stain Color Prediction

Testing steps — Finish Genius, Color Matching

Written for the people testing this at AWFI · Companion to “Automated Formulation & Wood.pdf”, in the same folder
Covers Finish Genius **Formulation** → **Color Matching**, plus the two other screens added alongside it.

What you are testing, and what you are not

The requirements describe a camera-based stain prediction system. What is in the app is the part that is software: a library of measured samples, matching a wanted colour against them, reading colour off a photograph, and showing a board as a stain should leave it.

Not in the app, so do not test for it: camera control and RAW files (you upload an ordinary photo), a trained model, and composing a brand-new formula for a colour nobody has mixed. Those need the sample library to exist first, or hardware. Everything the app answers traces to samples somebody measured — that is the point of these tests.

Before you start

1. Start the app (`restart-local.bat`) and sign in.
2. Pick the group you want to test in, at the top of the screen. Samples belong to a group, so record and search in the same one.
3. Open **Formulation** → **Color Matching**.
4. Nothing else is needed for tests 1–3 and 6–8. Test 4 needs a photo; test 5 needs a photo of bare wood. A spectrophotometer and a ColorChecker chart make test 4 meaningful but are not required to run it.

On Production the colour tables are created the first time the screen is opened. If your login cannot create tables, run `database/FG_ColorSamples.sql` once first.

Test 1 — Record three samples

This is the foundation: everything else is answered from these rows. On the **Sample Library** tab, press **New Sample** and enter each row below. Leave every other field empty.

Field	Sample 1	Sample 2	Sample 3
Species	Maple	Maple	Maple
Sanded to (grit)	180	180	180
Unfinished wood L* a* b*	72.4 / 8.1 / 24.6	72.4 / 8.1 / 24.6	72.4 / 8.1 / 24.6
Formula name	AWFI WB-042	AWFI WB-042	AWFI WB-042
Strength (%)	6	8	10
Applied by	Spray	Spray	Spray
Measured L* a* b*	64.9 / 10.8 / 28.2	61.2 / 11.4 / 29.7	58.7 / 12.1 / 27.3
Read from	Spectrophotometer	Spectrophotometer	Spectrophotometer

Expected: each save says “Colour sample saved.” The header then reads **3 samples, 1 wood, 1 formula, 3 device readings**, and each row shows a brown swatch with its L*/a*/b* beside it.

PRO

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admin@finishgenius.local

Home / Color Matching

Color Matching

Measured samples, the formula that gets closest to a colour, and what it should look like on the wood.

SAMPLES

4

Every prediction rests on these

WOODS

2

Species recorded

FORMULAS

2

With at least one sample

DEVICE READINGS

4

All read on a spectrophotometer

Sample Library

Match a Colour

Stain Preview

COLOUR

WOOD

FORMULA

STRENGTH

APPLIED

COATS

TOPCOAT

READ FROM

MEASURED

ACTIONS

55.1 / 10.7 / 25.9

Oak

AWFI WB-101

8%

Spray

1

Water-based urethane

Spectrophotometer

—

58.7 / 12.1 / 27.3

Maple

AWFI WB-042

10%

Spray

1

Water-based urethane

Spectrophotometer

—

61.2 / 11.4 / 29.7

Maple

AWFI WB-042

8%

Spray

1

Water-based urethane

Spectrophotometer

—

64.9 / 10.8 / 28.2

Maple

AWFI WB-042

6%

Spray

1

Water-based urethane

Spectrophotometer

—

Showing 1 to 4 of 4 entries

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Sample Library after recording the samples (a fourth, on Oak, was added for the later tests).

New colour sample

The wood

Species * Sanded to (grit) Measured on
 Maple, Oak, Cherry... mm/dd/yyyy

Grain Porosity Growth rings Moisture (%)
 — — —

Existing finish
 Anything already on the wood — leave empty for bare timber.

Unfinished wood colour (L* a* b*)
 L* a* b*

Optional, and worth doing: it lets the same formula be predicted on a different board.

The stain

Formula * Formula name *
 Select a formula

Pick one of this group's formulas, or type a name below.

Strength (%) Applied by Coats Wet film (mils)
 Spray

Spray gun Gun pressure (psi) Drying conditions
 Kremlin, Graco... e.g. 22 °C, 45% RH, overnight.

Flash (minutes) Sealer Topcoat Sheen (%)

The finished colour

Cancel Save

The New Sample form: the wood, the stain, how it was applied, and the finished colour.

Test 1b — The form refuses nonsense

1. Open **New Sample** and type `asasa` into **Sanded to (grit)**.
2. Type `150` into the first **Measured L*** box.

Expected: the grit box shows “**Numbers only.**”, L* shows “**Must be 100 or less.**”, and **Save** stays **greyed out** until both are sensible. Nothing bad reaches the database quietly.

Test 2 — Match a colour (the main test)

1. Go to the **Match a Colour** tab.
2. Enter the target **L* 63, a* 11.1, b* 29**.
3. Press **Find the closest formulas**.

Expected: AWF1 WB-042 at the top with **ΔE 0.05 · Not visible**, and — the important part — **“mix at 7%”**. Nobody mixed 7%: the app read that strength off the line between your 6% and 8% samples. The line underneath says what it stands on: *“3 recorded samples; 3 read on a spectrophotometer; strength read between two measured points.”*

A target between two measured strengths: the answer is a strength, not just the nearest sample.

Test 2b — The board you are actually matching on

Fill in **This board's unfinished colour** with **65 / 9 / 22** (a darker board than the samples) and search again.

Expected: the predicted colour and ΔE change, and the basis line adds **“corrected for this board's own colour”**. The app applies the stain's measured *shift* to your board instead of assuming the same wood.

Test 3 — It says when it does not know

These two checks are what tell you the answers in test 2 can be trusted.

1. Search for a colour nothing comes near: **L* 30, a* 40, b* 10**.

2. Go to **Stain Preview**, and pick a formula that has no samples (or invent a name in a sample search).

Expected: (1) the results are graded **“Different colour”** in red with low confidence — no invented match. (2) The preview reports **“No sample has ever been recorded for this formula”** instead of drawing something.

A colour nothing was recorded near: graded honestly rather than matched.

Test 4 — Read a colour from a photograph

A photo is not a measurement unless something of known colour is in the shot. Test both ways round.

1. Photograph a board with a **24-patch ColorChecker** (or a grey card) lying next to it, under even light.
2. On **Match a Colour**, open **Read the target off a photo** and choose the picture.
3. Press **Find the areas**. The board should be boxed for you.
4. Press **Mark the ColorChecker** and drag a box tightly around the chart. Then **Read the colour**.
5. Now repeat with no chart marked, to see the difference.

Expected: with the chart marked it reports “**Corrected against all 24 chart patches: the photo was ΔE 9 out, now 2.2.**” and fills the target boxes. Without it, the reading still appears but is marked **uncalibrated**, with a warning not to file it as a measurement. The two readings should differ — that difference is the room’s lighting.

The screenshot displays the AWFI Pro software interface. On the left is a dark sidebar with navigation menus: OPERATIONS (Dashboard, My Work, Work Instructions), PROCESSES (Process Steps, Process Schedules, Material Quantities, Pricing), FORMULATION (Equipment & Materials, Formulas, Color Matching, Photo Gallery), and ADMINISTRATION (Groups, Users, Material Categories, Sub Step Setup, Import, System Settings). The 'Color Matching' option is highlighted in orange. The main panel shows the 'Read the target off a photo' workflow. At the top, it displays color values: 59.75, 10.47, 29.74. Below this, there are dropdowns for 'Wood' (Any wood) and 'Topcoat' (Any topcoat). A section for 'This board's unfinished colour (optional)' includes L*, a*, and b* input fields. An orange button labeled 'Find the closest formulas' is present. The 'Read the target off a photo' section includes a 'Choose another photo' button and a note about fixed camera and lighting. Below this are three buttons: 'Mark the wood', 'Mark the ColorChecker', and 'Mark a grey card'. The 'Find the areas' button is active, showing a preview of a wood sample with an orange box and a ColorChecker chart with a green box. The bottom section shows the results: 'Wood found. No ColorChecker was read: mark it yourself if it is in the shot.' Below this is a small color swatch and the text: 'L* 59.75 a* 10.47 b* 29.74. Corrected against all 24 chart patches: the photo was ΔE 9 out, now 2.2.'

Wood boxed in orange, chart in green. Here the photo was ΔE 9 out and corrected to 2.2.

Worth knowing: if “Find the areas” does not box the chart, that is deliberate — it tries to read the 24 patches first and stays quiet rather than hand you a chart it could not fit. Mark it by hand and carry on. A large correction figure (over about ΔE 12) is the app telling you the lighting was poor, not that it failed.

Test 5 — Stain preview

1. Open the **Stain Preview** tab and choose a photo of bare wood.
2. Press **Find the wood**, or drag a box over the board yourself.
3. Choose formula **AWFI WB-042**, strength **9**, and press **Show it stained**.

Expected: the board comes back darker with its grain intact, and underneath: **“Predicted L* 59.95 a* 11.75 b* 28.5 · read between the 8% and 10% samples”** with a confidence figure. A 12-megapixel photo takes about a second.

The screenshot shows the AWFI Stain Preview software interface. The top navigation bar includes a 'Group' dropdown set to 'AWFI', a 'Development' button, and links for 'Help & support', 'DPM Center', and a user profile 'Hello, System' with email 'admin@finishgenius.local'. The left sidebar is divided into 'OPERATIONS' (Dashboard, My Work, Work Instructions), 'PROCESSES' (Process Steps, Process Schedules, Material Quantities, Pricing), 'FORMULATION' (Equipment & Materials, Formulas, Color Matching - highlighted), 'Photo Gallery', and 'ADMINISTRATION' (Groups, Users, Material Categories, Sub Step Setup, Import, System Settings). The main content area has tabs for 'Sample Library', 'Match a Colour', and 'Stain Preview' (active). The 'Stain Preview' tab is split into two panels. The left panel, 'The board and the stain', contains a 'Photo of the unfinished wood' section with a 'Choose another photo' button, a 'Drag a box over the bare wood' section with a 'Find the wood' button, a photo of a wood board with a selection box, and a 'Formula' dropdown set to 'AWFI WB-042'. Below this are 'Strength (%)' set to '9' and 'Wood' set to 'Any recorded', with a 'Measured: 6, 8, 10' note and a 'Show it stained' button. The right panel, 'Predicted result', shows a wood board with a selection box and a color swatch. Below the swatch, it displays 'Predicted L* 59.95 a* 11.75 b* 28.5' and '50% confidence - read between the 8% and 10% samples, on every wood recorded for it. From 3 samples.' A warning box at the bottom states: 'The colour is predicted from measured samples; the grain, gloss and any blotching are this photograph's, not a prediction. Treat it as a guide for the customer, and prove it on a real sample board before finishing.' The bottom of the interface shows a 'Collapse' button and a copyright notice '© AWFI 2026'.

The colour is predicted from the samples; the grain is the photograph's own.

Test 6 — The recipe is kept with the sample

1. Create a sample and, in **Formula**, pick one of the group's real formulas instead of typing a name.
2. Save it, then open it again with the pencil.

Expected: a “**Recipe as recorded**” list appears, with every colorant, its grams and its percentage of the batch. It is copied onto the sample, so editing that formula later does not change what this sample says was on the wood. The same percentages appear under each recommendation on Match a Colour.

Test 7 — The conditions that change a colour

Record a sample filling in **grain direction, porosity, growth rings, existing finish, moisture, spray gun, gun pressure and drying conditions**, then reopen it.

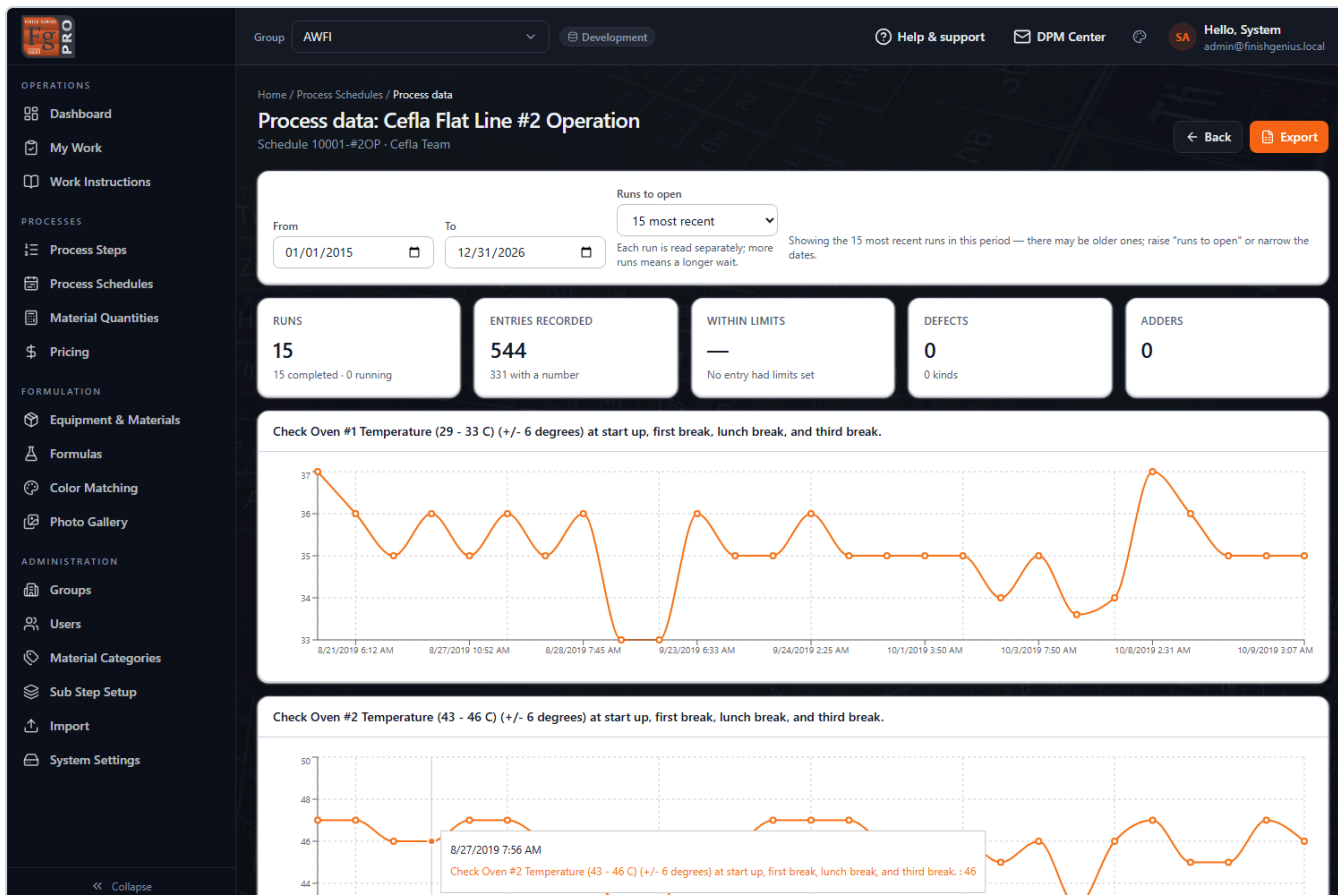
Expected: everything comes back as entered. These are the conditions the requirements list as changing the result, which is why they are recorded next to the colour.

Test 8 — Two other new screens

Process data (Process Schedules → the activity icon)

1. Open **Process Schedules** and press the activity icon on a schedule that has been run.
2. Set a date range that covers those runs.

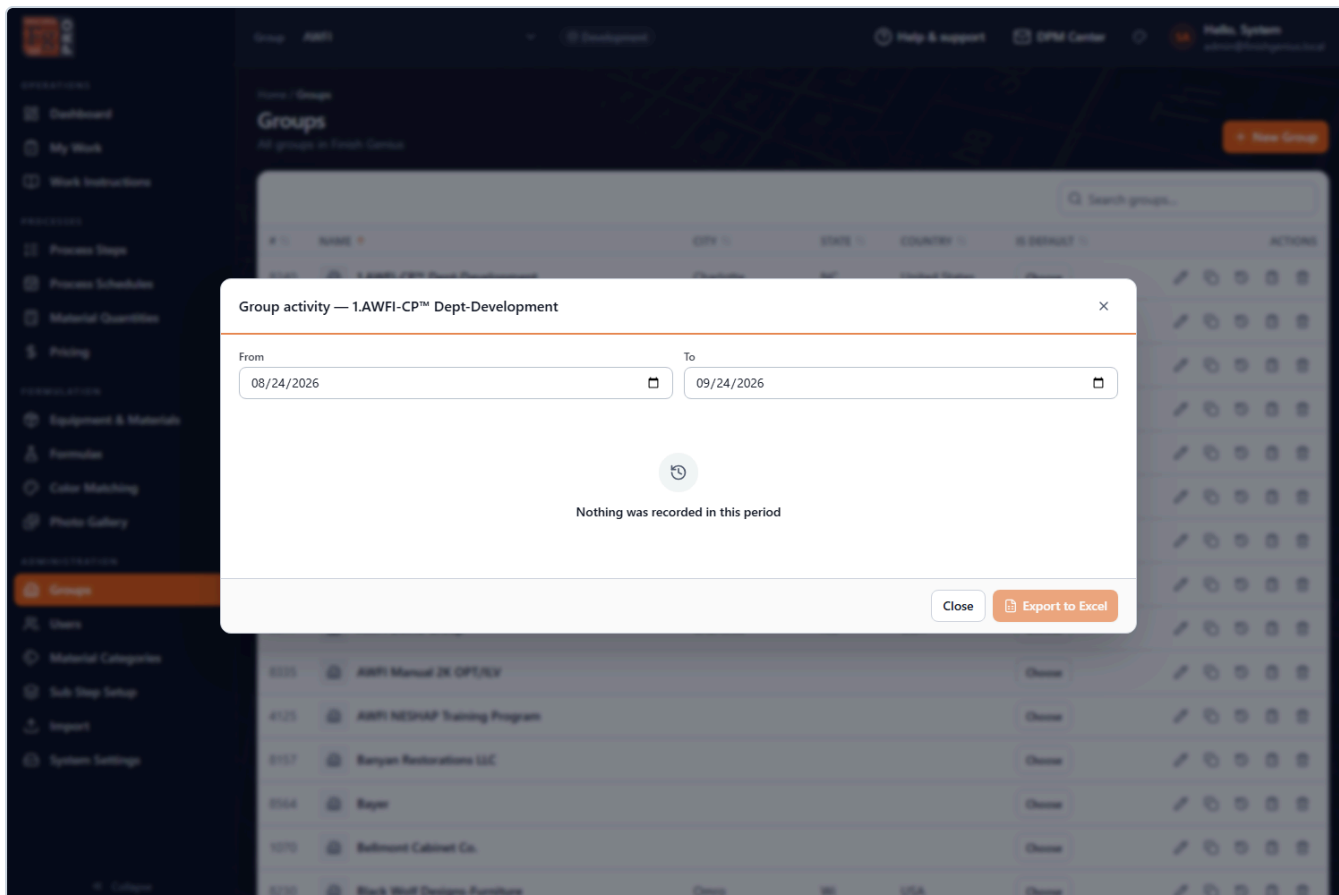
Expected: totals across the runs, a line chart for every checklist entry that records a number (with the schedule's limits drawn on it), and the entries, defects and adders as tables. **Export** gives a four-sheet workbook. This replaces the old site's Graphs, quality-control charts and Process Data Details.



Oven temperatures across 15 runs of one schedule, from what operators recorded.

Group activity (Groups → the clipboard icon)

Expected: everything that happened in the group over a date range, searchable, with **Export to Excel** for the whole period. This replaces the old site's "Group Download".



A group's whole audit trail for a period, readable before it is exported.

Getting useful results afterwards

Once the tests pass, the value depends entirely on the library you build. The requirements suggest starting with **3–5 woods × 20–30 formulas × several strengths**, applied the way the shop really applies them. Three habits matter:

1. **Measure the bare board too**, not just the finished sample. It is one extra reading, and it is what lets a formula be predicted on a board it was never tried on.
2. **Record at least two strengths** of any formula you care about. That is what turns “nearest recorded colour” into “mix at 7%”.
3. **Record what you actually got** after mixing to a recommendation. That is how the next search gets better.

Result sheet

#	Test	Pass?	Notes
1	Three samples recorded; header counts correct		
1b	Form refuses letters and out-of-range numbers		
2	Match returns WB-042, ΔE 0.05, mix at 7%		
2b	Board colour changes the prediction and is said so		
3	Far colour graded “Different colour”; unknown formula refused		
4	Photo reading calibrated by chart; uncalibrated marked as such		
5	Stain preview renders with the predicted colour and basis		
6	Recipe captured with the sample and shown with matches		
7	Wood and application conditions saved and reloaded		
8	Process data charts and export; group activity export		

Tester: _____ Date: _____ Database tested: Development / Production